

An Executable Decision-Rights Matrix and an Expert-Approved Logic Database

Datalog with stratified negation for deterministic, explainable
and attributable decisions in human-centred agentic systems

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ACM Europe School on Responsible AI

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At volume, this is unaffordable — so it is skipped,
or done at a rate that makes it ceremonial.

And a reviewer shown a recommendation without its reasoning
can do little besides agree.

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Review the **rule**, not the case.

A judgement made once is recorded, attributed, and reused.

Contribution 1: an executable decision-rights matrix

Our group's matrix assigns one of four tiers of AI authority from **harm**, **reversibility**, **rights** and **evidence**.

human_only	AI may not decide
human_approval	a person approves before it takes effect
ai_with_oversight	bounded, logged, reversible
ai_recommends	propose; do not act

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Most such artefacts live in a document.

This one decides, in nine lines, in a module that **imports nothing**.

Two implementations, tested for agreement

In Python on the server; in JavaScript in the page, so the sliders respond without a round trip.

- Two copies of the rule allocating oversight can drift with **nothing failing**.
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A test extracts the classifier *from the shipped page*, runs it under Node, and compares **all 1,000 input combinations**.

Mutating one threshold: twenty disagreements, each named.

The gate asks one question

Has a person already ruled on this logic?

Tier	Approved rule fires	Nothing fires
human_only	proceeds	waits
human_approval	proceeds	waits
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The tier sets what is at stake — not whether a person is asked.

Why approved logic proceeds even at the top tier

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- Asking again for every matching case is how oversight becomes ceremony.
- A proportion is audited anyway — 50 % at `human_only`, 5 % routine, **never zero**.
- Sampling is deterministic on the case: an auditor can recompute it.
- Any rule can be retired, which enumerates what it already decided.

Expert review of prepared decisions

The matrix says the AI *prepares* the action and a person rules on it.

- So the answer **is** written — withheld from the person waiting, shown to the expert.
- In an **editable** field. An expert who may only accept or refuse is not reviewing, they are signing.
- What is sent is attributed to them, and says whether they revised it.

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Nothing is said to the person waiting in the meantime.

A holding answer gesturing at the outcome *would be* the decision.

Their judgement becomes logic — once

An expert rules. The system drafts the clause their ruling implies, checks it fires on the case's own facts, and offers it **in the same interaction**:

```
| not_evict(c) ← in_arrears(c), disputes_debt(c),  
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Not approved automatically.

They ruled on an *answer*; this is a generalisation of that answer, and may be wider than they meant. One further click, with the clause in front of them.

Contribution 2: the logic database as formal substrate

Recording a person's judgement as a rule is only honest if the rule behaves.

- **Determinism** — function-free Horn clauses, a terminating fixpoint, snapshot matching. Same facts, same rules, same answer.
- **Explainability** — forward chaining, so the derivation *exists*. The proof is a transcript, not a story told afterwards.
- **Transparency** — every rule carries its approver and its record; every decision its facts, its rules and its tier; history is append-only.
- **Formal footing** — stratified negation, canonical clause identity, safety checked at construction.

Evidence that stays honest

```
will_get_wet(C) ← going_out(C), raining(C),  
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- Umbrella recorded **absent** → fires.
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Facts come from a model reading free text. Under a closed world, silence becomes evidence against a person one round later.

We shipped that defect twice in a predecessor.

Near-miss detection and generalisation

What almost fired. A clause needing three conditions with two met is not a miss — it is a question, and a precise one:

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merge would do. Two clauses alike but for one condition merge to their shared core — replayed over every stored case first, and shown with what it keeps, newly decides, and contradicts.

Never applied automatically: a generalisation fires on strictly more cases than its sources.

Failures encountered during development

- **A greeting was classified.** “Hi” → *human approval required*: confidence 0, and the rule reads confidence < 70.
- **A gate that overfitted its own prompt.** 6/6 handover on “toner”, 0/6 on “envelopes” — the difference being which word was in our example list.
- **The model was the limit, not the prompt.** Four rounds of rewording; then we measured. Fact extraction: Haiku **0/4**, Sonnet **4/4**.
- **Conclusions stored as facts.** The extractor was offered rule *heads* as vocabulary and wrote one down as evidence.
- **Silent no-ops.** Three times an edit failed to match and left code that looked right and did nothing.

Coverage results

	Run A	Run B
Decided from approved logic	2 / 12	3 / 12
Parked for a person	5 / 12	4 / 12
Coverage	18 %	30 %
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Twelve cases, empty database, every proposed rule approved.
We report the range, not the better number.

Why it is not higher

The vocabulary the database built for itself:

reorder_toner	reorder_paper
toner_stock_low	paper_stock_low

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- Coverage appears where the entity stayed *out* of the predicate — `in_arrears`, `debt_disputed`.
- It fails where the entity went *into* the name.

`reorder(C, X) ← stock_low(C, X).`

The engine is first-order and can express this. Nothing proposes it:
generalisation can widen a rule, it cannot repair a vocabulary.

What we have not measured

- **No expert has used this.** Every approval in those runs was granted by a script.
- **Two runs is not a measurement.** 18 versus 30 per cent on identical input.
- **Top-tier conduct is a prompt, not a mechanism.** It held under direct pressure — three trials is an anecdote.
- **The matrix has defects we did not fix.** Reversibility does nothing at harm 4–7; confidence overrides harm entirely. We report them rather than quietly amend our group's artefact.

Human-centred is not a person in every loop.

It is choosing the unit of judgement
so that judgement scales.

A matrix that allocates oversight and never sees the answer.

A logic database that produces the answer and never allocates oversight.

A gate whose only question is whether a person has already ruled.

A fifth of cases decided the same way every time, with a derivation
attached and a named person behind the rule.